Guidelines

- Kindly use the given template for submitting your prototype(Make a copy of the template)
- One team is only required to submit one project.
- The ideal size of the presentation should not be more than 10 slides.
- You are welcome to add as many POCs and design concepts to support your project.
- The project should be feasible and the team members should be capable enough, to come up with the fully functioning prototype of the same idea, if required. In case of queries, kindly reach out to us at codeforge@mrem.ac.in

Team Details

a. Team Name: SPIRIT
b. Tittle: Self-Sovereign Academic Credential Verification System
c. Problem Statement: Fake academic certificates and time-consuming manual verification processes create trust issues in education and hiring systems.

Brief about your solution

We propose a blockchain-based system using Algorand to securely store and verify academic certificates. Universities upload certificates as unique hashes, students can access and share them, and recruiters can verify them instantly using a QR code. This ensures certificates cannot be faked or changed and makes verification fast and reliable

Opportunities

a. How different is it from any of the other existing

- Uses blockchain instead of centralized storage
- Ensures tamper-proof and secure certificates

b. ideas? How will it be able to solve the problem?

- Eliminates fake certificates Provides instant and reliable verification

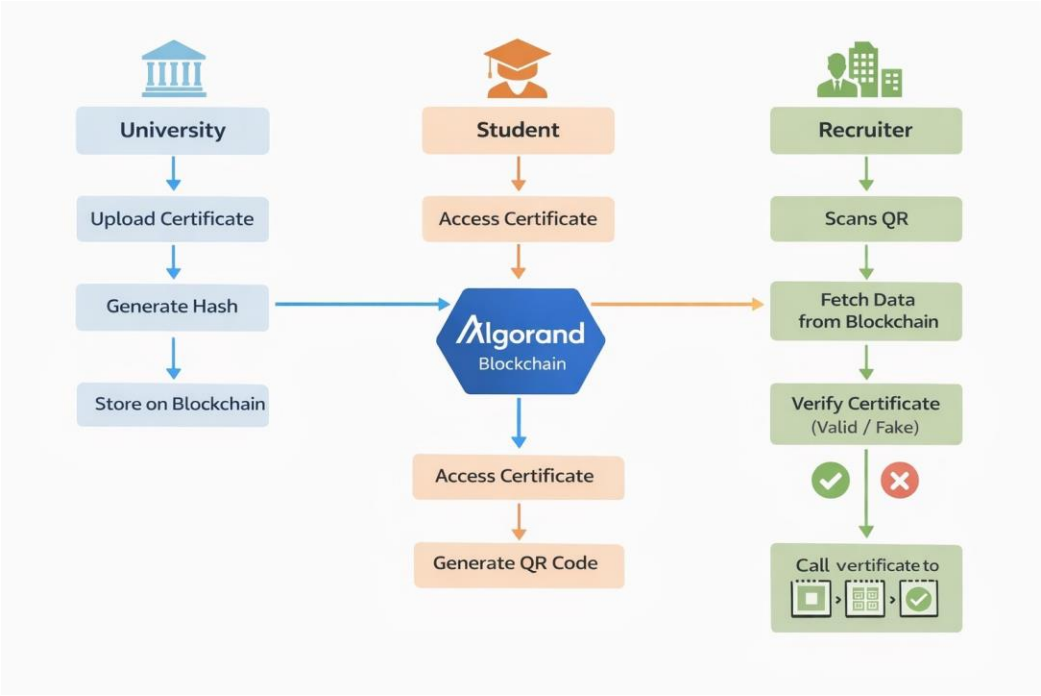
c. USP of the proposed solution.

- Self-sovereign ownership (student controls data)
- QR-based real-time verification Transparent and decentralized system

List of features offered by the solution

Secure certificate issuance
Blockchain-based hash storage
Student access and sharing
QR code-based verification
Instant authenticity check
Certificate revocation option

Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution (optional)

Login Page (University / Student /Recruiter)
University Dashboard (Upload certificate)
Student Dashboard (View & share certificate)
Verification Page (QR scan result)

Architecture diagram of the proposed solution

Technologies to be used in the solution

Blockchain: Algorand
Smart Contracts: PyTeal / TEAL
Frontend: React
Backend: Node.js / Flask
Wallet: Pera Wallet
Storage: IPFS / Firebase
QR Code Generator